



THE KING'S SCHOOL

2003
Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 84

- Attempt Questions 1-7
- All questions are of equal value

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Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) $P(5,7)$ divides the interval AB externally in the ratio $m:n$

If $A = (-1, -5)$ and $B = (0, -3)$, find $m:n$. 2

- (b) Find the derivative of $\tan^{-1}(1+x^2)$ 2

- (c) In how many ways can 8 rowers be divided into two groups of 5 and 3 rowers? 2

- (d) Find the acute angle between the lines $y = 2x + 1$ and $y = 7 - 3x$. 2

- (e) $x + 1$ is a factor of the polynomial

$$P(x) = x^{2n+1} - x^{2n} + b, \quad n \text{ a positive integer}$$

Find the value of b . 2

- (f) Evaluate $\sum_{n=1}^9 \left(\frac{1}{n} - \frac{1}{n+1} \right)$ 2

Question 2 (12 marks) Use a SEPARATE writing booklet.

(a) Show that the function $f(x) = 5x - \sin 4x - 12$

increases for all values of x

3

(b) (i) Find $R > 0$ and α , $0 < \alpha < \frac{\pi}{2}$, so that

$$R \sin(x - \alpha) = \sin x - \sqrt{3} \cos x$$

2

(ii) Solve $\sin x - \sqrt{3} \cos x = \sqrt{2}$, $0 < x < 2\pi$, exactly

2

(c) (i) Use the substitution $u = 4 - x^2$ to evaluate

$$\int_0^{\sqrt{3}} \frac{x}{\sqrt{4 - x^2}} dx$$

3

(ii) Evaluate $\int_0^{\sqrt{3}} \frac{4 - x}{\sqrt{4 - x^2}} dx$

2

Question 3 (12 marks) Use a SEPARATE writing booklet.

- (a) The function $f(x) = e^{x^2} - x - 3$ has a zero near $x = 1.2$.

Use Newton's Method once to find a two decimal place approximation to this zero.

3

- (b) (i) Write $\sin 2A$ in terms of t , where $t = \tan A$

1

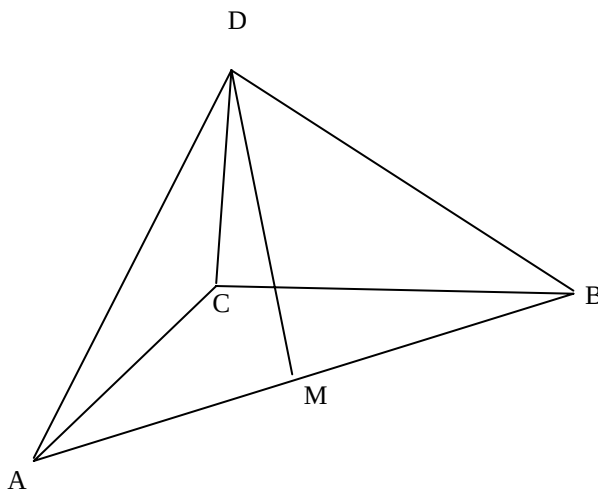
- (ii) Prove the identity $\operatorname{cosec} 2A - 3\cot 2A = 2\tan A - \cot A$

3

- (c) A, B, C are three points in a horizontal plane and M is the mid-point of AB. CD is at right-angles to the horizontal plane ABC.

The length of AB = 20 metres.

The angles of elevation from A, M, B to D are 30° , 45° , 30° , respectively.



- (i) Copy the diagram into your booklet and include on it the information given.

1

- (ii) If $CD = x$, show that $AC = \sqrt{3}x$

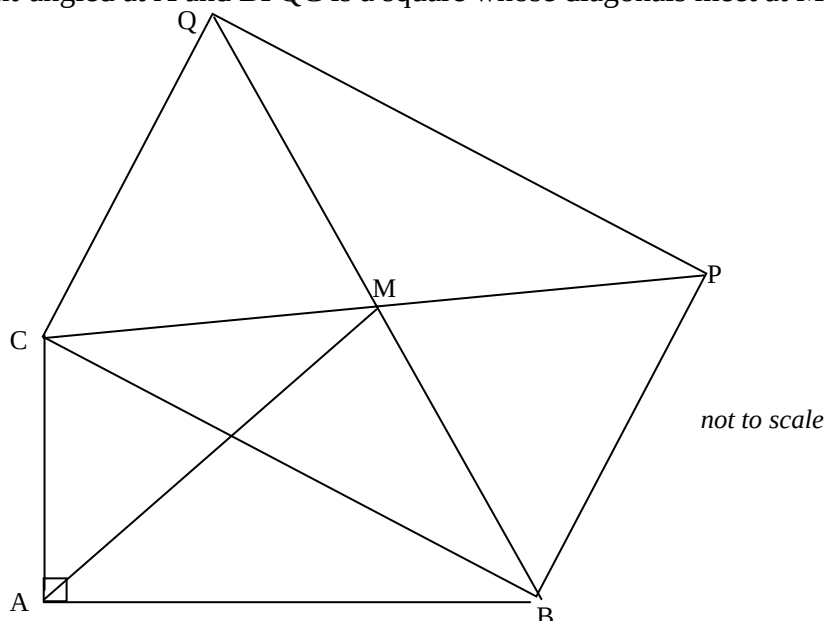
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- (iii) Find the exact value of x .

3

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) $\triangle ABC$ is right-angled at A and BPQC is a square whose diagonals meet at M.



- (i) Copy the diagram into your booklet.
- (ii) Prove that ABMC is a cyclic quadrilateral. 2
- (iii) Prove that MA bisects $\angle BAC$ 2
- (b) A particle is moving in simple harmonic motion on the x -axis. Its displacement, x metres, at any time t seconds, where $t \geq 0$, is given by $x = 10 \cos nt$, n a constant.
- (i) Show that the particle is initially at rest at $x = 10$. 2
- (ii) The period of the motion is T seconds and after $\frac{T}{3}$ seconds the particle is at position $x = b$.
Find the value of b . 2
- (iii) The speed at $x = b$ is $20\sqrt{3}$ m/s. Find the period of the motion. 2
- (c) Simplify $\binom{n}{3} \div \binom{n-1}{2}$, $n \geq 3$ 2

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) $P(2ap, ap^2)$, $Q(2aq, aq^2)$ are the end points of a focal chord of the parabola $x^2 = 4ay$.

- (i) Show that the equation of the chord PQ is

$$y - \frac{1}{2}(p+q)x + apq = 0 \quad 2$$

- (ii) Deduce that $q = -\frac{1}{p}$ 1

- (iii) Show that $PQ = 2a + a\left(p^2 + \frac{1}{p^2}\right)$ 2

- (iv) A circle is drawn with PQ as its diameter.

Prove that the directrix is a tangent to this circle. 3

- (b) A cube is expanding in such a manner that it maintains its cubic shape. Initially, each edge is 10cm and the surface area is expanding at a constant rate of $12.6 \text{ cm}^2/\text{s}$.

- (i) Find an expression in terms of t for the surface area of the cube after t seconds. 1

- (ii) Hence, or otherwise, find the rate at which the volume of the cube is increasing after 10 seconds. 3

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Prove by mathematical induction for $n \geq 0$ that $E(n) = 9^{n+2} - 4^n$ is a multiple of 5. 3

- (b) (i) With the aid of a diagram, or otherwise,

solve $(x^2 - 1)(x^2 - 4) \leq 0$ 2

- (ii) A particle is moving along the x -axis with its acceleration at position x given by

$$\ddot{x} = 10x - 4x^3$$

When $x = \sqrt{2}$ its velocity $v = 2$

- (α) Prove that the expression $\frac{1}{2}v^2 + x^4 - 5x^2$ is a constant for the motion and find this constant.

(v is its velocity at position x) 2

- (β) Describe the motion. 2

- (c) A random sample of 10 people is made. Assuming that either sex is equally likely, find the probability that

- (i) there is an equal number of each sex 1

- (ii) there are more females 2

Give your answers correct to three decimal places.

Question 7 (12 marks) Use a SEPARATE writing booklet.

(a) Solve $\frac{-2x}{x+1} > 0$ 2

(b) Consider the curve $y = \ln\left(\frac{-2x}{x+1}\right)$

(i) Use (a) to explain why

$$\ln\left(\frac{-2x}{x+1}\right) = \ln(-2x) - \ln(x+1)$$
 1

(ii) Show that the curve has no stationary points. 2

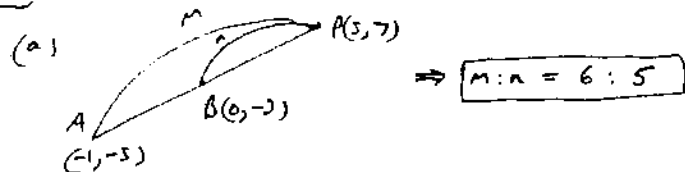
(iii) Sketch the curve, showing the x -intercept. 2

(iv) Find the inverse function of $y = \ln\left(\frac{-2x}{x+1}\right)$,
expressing your answer with y as subject. 3

(v) Find the area of the region bounded by $y = \ln\left(\frac{-2x}{x+1}\right)$ and the y -axis
and the lines $y = 0$ and $y = 2$ 2

End of Paper

Ques 1



(b) $\frac{d}{dx} \tan^{-1}(1+x^2) = \frac{1}{1+(1+x^2)^2} \times 2x = \frac{2x}{1+(1+x^2)^2}$

(c) $\binom{8}{5}$ or, of course, $\binom{8}{3} = 56$

(d) gradients of lines are 2 and -3
 $\therefore \tan \angle = \left| \frac{2 - (-3)}{1 + 2(-3)} \right| = \frac{5}{5} = 1$
 $\therefore$ acute angle is 45°

(e) $P(-1) = 0 \Rightarrow (-1)^{2n+1} - (-1)^{2n} + b = 0$
 $\therefore -1 - 1 + b = 0 \therefore b = 2$

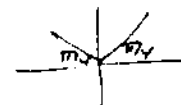
(f) $\sum_{n=1}^9 \left(\frac{1}{n} - \frac{1}{n+1} \right) = \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{9} - \frac{1}{10} \right)$
 $= 1 - \frac{1}{10}$
 $= \frac{9}{10}$

Ques 2

(a) $f'(x) = 5 - 4 \cos 4x$
 $\geq 5 - 4(1)$ since $-1 \leq \cos 4x \leq 1$
 $\Rightarrow f'(x) \geq 1 > 0 \quad \forall x$
 $\therefore f(x)$ increases $\forall x$

(b) (i) $R \sin(x-\alpha) = R \cos \alpha \sin x - R \sin \alpha \cos x$
 $= \sin x - \sqrt{3} \cos x$
 $\Rightarrow R \cos \alpha = 1$
 $R \sin \alpha = \sqrt{3}$
 $\therefore \tan \alpha = \sqrt{3}, \alpha = \frac{\pi}{3}$
 and $R = \sqrt{1^2 + \sqrt{3}^2} = 2$

(ii) From (i), $2 \sin(x - \frac{\pi}{3}) = \sqrt{2}$
 $\therefore \sin(x - \frac{\pi}{3}) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$
 $\therefore x - \frac{\pi}{3} = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$
 $\therefore x = \frac{7\pi}{12} \text{ or } \frac{13\pi}{12}$



(c) (i) $u = 4 - x^2$
 $\frac{du}{dx} = -2x$ or $du = -2x dx$
 $x=0, u=4$
 $x=\sqrt{3}, u=1$
 $\therefore I = -\frac{1}{2} \int_4^1 \frac{du}{\sqrt{u}} = \frac{1}{2} \int_1^4 u^{-\frac{1}{2}} du$
 $= \frac{1}{2} \cdot 2 \left[u^{\frac{1}{2}} \right]_1^4$
 $= 2 - 1 = 1$

(ii) $I = \int_0^{\sqrt{3}} \frac{4}{\sqrt{4-x^2}} - \frac{x}{\sqrt{4-x^2}} dx = 4 \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} - 1$, from (i)
 $= 4 \cdot \frac{\pi}{3} - 1$

(a) $f(x) = 2x e^{x^2} - 1$

$$\therefore x_1 = 1.2 - \frac{e^{1.44} - 1.2 - 3}{2.4e^{1.44} - 1} = 1.1977 \dots$$

$\therefore$ two decimal approxⁿ = 1.20

Ans] (i) $\sin 2A = \frac{2t}{1+t^2}$

(ii) put $x = \tan A$,

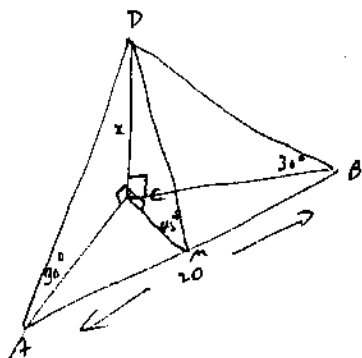
the $\operatorname{cosec} 2A - 3 \cot 2A = \frac{1+t^2}{2t} - 3 \cdot \frac{1-t^2}{2t}$

$$= \frac{1+t^2 - 3 + 3t^2}{2t}$$

$$= \frac{4t^2 - 2}{2t}$$

$$= 2t - \frac{1}{t}$$

$$= 2 \tan A - \cot A$$

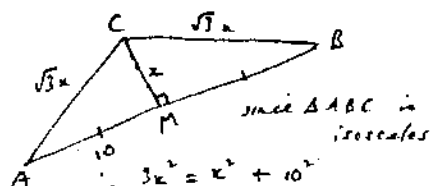


(ii) $I \triangle ACD$,

$$\tan 30^\circ = \frac{x}{AC} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow AC = \sqrt{3} \times$$

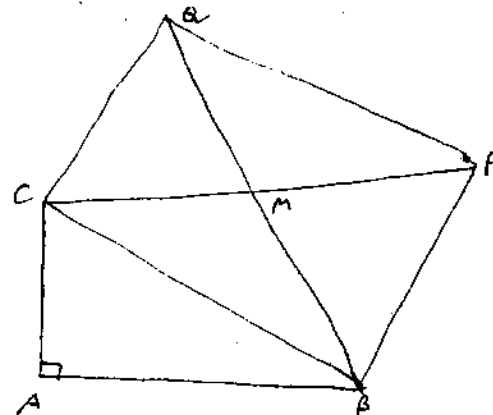
(iii) From (ii), we have



$$\begin{aligned} 3x^2 &= x^2 + 10^2 \\ 2x^2 &= 100 \\ x^2 &= 50 \end{aligned}$$

$$\therefore x = \sqrt{50} = 5\sqrt{2}$$

(i)



(ii) $\angle CMB = 90^\circ$, the diagonals of a square meet at right angles.

$\therefore \angle \diamond ABMC, \angle A + \angle M = 180^\circ$

$\Rightarrow \diamond ABMC$ is cyclic, opposite angles are supplementary

(iii) $MC = MB$, equal diagonals in a square bisect each other.

$\therefore \angle CAM = \angle BAM$, $\angle$ s at the circumference of a circle standing on equal arcs.

MA directs LBAC

(4) (i) $t = 0, \quad x = 10 \cos 0 = 10$

$$\dot{x} = -10x \sin t = -10x \sin 0 \text{ at } t=0$$

$$= 0$$

or particle is initially at rest at $x=0$

(ii) $T = \frac{2\pi}{\omega}$ $\therefore \omega = \frac{2\pi}{T}$

$$\therefore b = 10 \cos\left(\frac{2\pi}{T} \cdot \frac{T}{3}\right) = 10 \cos\left(\frac{2\pi}{3}\right)$$

$$\therefore b = 10 \left(-\frac{1}{2}\right) = -5$$

$$(iii) \quad \ddot{x} = -10a \sin \omega t = -10 \cdot \frac{2\pi}{T} \sin\left(\frac{2\pi t}{T}\right)$$

$$\therefore -20\sqrt{3} = -\frac{20\pi}{T} \sin\left(\frac{2\pi}{3}\right) = -\frac{20\pi}{T} \cdot \frac{\sqrt{3}}{2}$$

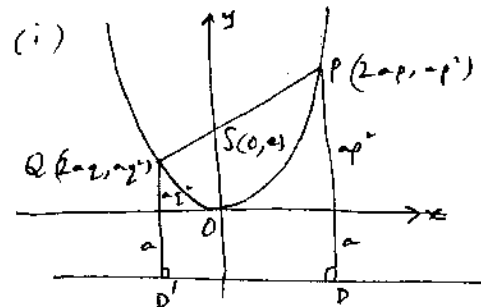
$$\therefore T = \frac{\pi}{2} \text{ (seconds)} \quad \therefore \text{period is } \frac{\pi}{2} \text{ s}$$

$$(C) \quad \binom{n}{3} \div \binom{n-1}{2} = \frac{n!}{(n-3)! 3!} \div \frac{(n-1)!}{(n-3)! 2!}$$

$$= \frac{n!}{(n-3)! 3!} \times \frac{(n-3)! 2!}{(n-1)!} = \frac{n}{3}$$

Q 5

(a) (i)



$$\begin{aligned} \text{gradient } PQ &= \frac{ap^2 - az^2}{2ap - 2az} \\ &= \frac{-p^2 + z^2}{2a(p-z)} \\ &= \frac{p+z}{2} \end{aligned}$$

$$\therefore \text{chord } PQ \text{ is } y - ap^2 = \frac{1}{2}(p+z)(x - 2ap)$$

$$\sim y - ap^2 = \frac{1}{2}(p+z)x - ap(p+z)$$

$$\Rightarrow y - \frac{1}{2}(p+z)x + apz = 0$$

(ii) Since $S(0, a)$ is on PQ , then

$$a - 0 + apz = 0$$

$$\therefore pz = -1 \quad \text{or } z = -\frac{1}{p}$$

(iii) $PQ = PS + SQ = PD + QD'$, focus-directrix defn (see diagram)

$$= a + ap^2 + a + az^2, \quad z = -\frac{1}{p}$$

$$= 2a + a\left(p^2 + \frac{1}{p^2}\right)$$

(iv) If PQ is a diameter, the radius is $a + \frac{a}{2}\left(p^2 + \frac{1}{p^2}\right)$,
from (iii)

$$\therefore \text{the centre is } \left(\frac{2ap + 2az}{2}, \frac{ap^2 + az^2}{2}\right)$$

$$= \left(a\left(p - \frac{1}{p}\right), \frac{a}{2}\left(p^2 + \frac{1}{p^2}\right)\right), \quad z = -\frac{1}{p}$$

$\therefore$ distance from centre to directrix $y = -a$ is

$$\frac{a}{2}\left(p^2 + \frac{1}{p^2}\right) + a = \text{radius}$$

$\therefore$ directrix is a tangent to the circle

(b) (i) $A = 6 \times 10^2 + 12.6t = 600 + 12.6t$

(ii) If an edge is x , $A = 6x^2$, $V = x^3$

$\therefore$ From (i), $6x^2 = 600 + 12.6t$

$x^2 = 100 + 2.1t$

$\therefore V = (100 + 2.1t)^{3/2}$

$\therefore$ So, $\frac{dV}{dt} = \frac{3}{2} (100 + 2.1t)^{1/2} (2.1)$

$= 3.15 \times \sqrt{121} \text{ cm}^3/\text{s}$ when $t=10$

$= 34.65 \text{ cm}^3/\text{s}$

Alternatively, using $A = 6x^2$, $V = x^3$, $\frac{dA}{dt} = 12.6$,

we have $\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$

$= \frac{dV}{dx} \cdot \frac{dx}{dA} \cdot \frac{dA}{dt}$

$= 3x^2 \cdot \frac{1}{12x} \cdot (12.6)$

$= 3.15x$

But, when $t=10$, $A = 6 \times 10^2 + 12.6 \times 10 = 726$

$\therefore 6x^2 = 726 \Rightarrow x = 11$

$\therefore \frac{dV}{dt} = 3.15 \times 11 \text{ cm}^3/\text{s} = 34.65 \text{ cm}^3/\text{s}$

Qn 6

(a) $E(0) = 9^2 - 4^0 = 80$ is a multiple of 5

$\therefore$ assume $E(n) = 9^{n+2} - 4^n = 5q$, q an integer, $n \geq 0$

Then, $E(n+1) = 9^{n+3} - 4^{n+1}$

$= 9(9^{n+2}) - 4^{n+1}$

$= 9(5q + 4^n) - 4^{n+1}$, using the assumption

$= 5(9q) + 4^n(9-4)$

$= 5(9q + 4^n)$ is a multiple of 5

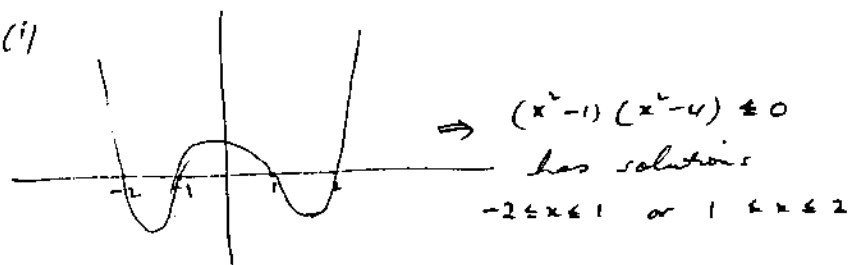
since $9q + 4^n$ is an integer

$\therefore$ if $E(n)$ is a multiple of 5, so is $E(n+1)$

But, $E(0)$ is a multiple of 5

$\therefore E(n)$ is a multiple of 5 by induction

(b) (i)



(ii) (L) $\frac{d(t v^2)}{dx} = 10x - 4x^3$

$\therefore \frac{1}{2} v^2 = 5x^2 - x^4 + C$, C a constant

i.e. $\frac{1}{2} v^2 + x^4 - 5x^2 = C$

When $x = \sqrt{2}$, $v = 2$

$\therefore 2 + 4 - 10 = C = -4$

(B) we have $\frac{1}{2}v^2 = 5x^2 - x^4 - 4$

$$\text{or } \frac{1}{2}v^2 = -(x^4 - 5x^2 + 4)$$

$$= -(x^2 - 1)(x^2 - 4)$$

$$\text{Now, } \frac{1}{2}v^2 \geq 0 \Rightarrow (x^2 - 1)(x^2 - 4) \leq 0$$

Using (b)(i) and when $x = \sqrt{2}$, $v = 2$, we have the particle oscillates between $x = 1$ and $x = 2$

(C) (i) $P(5 \text{ males, } 5 \text{ females}) = \binom{10}{5} \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^5$
 $= 0.246$, 3 d.p.

(ii) $P(\text{more females}) = P(\text{more males})$, since $P(M) = P(F) = \frac{1}{2}$
 $\therefore$ using (i), $P(\text{more females}) = \frac{1 - 0.246}{2} = 0.377$

Qn 7

(a) if $x+1 > 0$, then $-2x > 0$

$$\text{i.e. } x > -1$$

$$\text{i.e. } x < 0$$

$\therefore$ solution is $-1 < x < 0$

if $x < -1$, we'd have $x > 0 \Rightarrow$ no further solutions

$\therefore -1 < x < 0$

(b)(i) We need $\frac{-2x}{x+1} > 0$ and $-2x > 0$ and $x+1 > 0$.

All 3 inequalities are "true" from (a)

(ii) From (i), $y = \ln(2x) - \ln(x+1)$

$$\therefore \frac{dy}{dx} = \frac{-2}{-2x} - \frac{1}{x+1}$$

$$= \frac{1}{x} - \frac{1}{x+1} = \frac{x+1-x}{x(x+1)}$$

$$= \frac{1}{x(x+1)} \neq 0 \text{ for any } x$$

$\therefore$ there are no stationary points

(iii) Since $-1 < x < 0$, then $x(x+1) < 0$

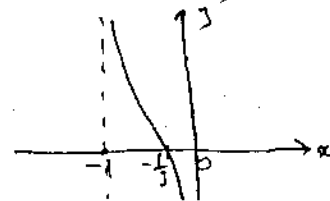
$$\therefore \text{From (ii)} \frac{dy}{dx} < 0 \text{ for } -1 < x < 0$$

i.e. curve is decreasing for $-1 < x < 0$

When $y = 0$, $\frac{-2x}{x+1} = 1$

$$\therefore -2x = x+1$$

$$\Rightarrow x \text{ intercept is } -\frac{1}{3}$$



(iv) Since curve is decreasing, the inverse function

$$\text{is } x = \ln\left(\frac{-2y}{y+1}\right)$$

$$\therefore \frac{-2y}{y+1} = e^x$$

$$-2y = e^x y + e^x$$

$$\therefore y(e^x + 2) = -e^x$$

$$\therefore \text{inverse function is } y = -\frac{e^x}{e^x + 2}$$

$$(v) A = \left| \int_0^2 x \, dy \right| = \int_0^2 \frac{e^y}{e^y + 2} \, dy, \text{ from (iv)}$$

$$= \left[\ln(e^y + 2) \right]_0^2$$

$$= \ln(e^2 + 2) - \ln 3 \quad \text{u}^2$$